MCA101P – PYTHON PROGRAMMING LAB

Lab Sheet-07: File Handling – Read and Write Operations

30 Python programs with sample outputs

Note: Each program is self-contained. It creates the sample file(s) it needs, so you can run each program separately. For Program 7, the sample output shows example user input.

Program 1: Create and write text into a file

Code:

```
# Create a file and write text into it
with open("sample.txt", "w") as file:
    file.write("Hello, Python!")

print("Data written successfully.")
```

Output:

```
Data written successfully.
```

Program 2: Read the entire content of a text file

Code:

```
# Read the complete file
with open("sample.txt", "w") as file:
    file.write("Hello, Python!")

with open("sample.txt", "r") as file:
    content = file.read()

print(content)
```

Output:

```
Hello, Python!
```

Program 3: Read a file line by line

Code:

```
# Read and display each line
with open("lines.txt", "w") as file:
    file.write("First line\\nSecond line\\nThird line")

with open("lines.txt", "r") as file:
    for line in file:
        print(line.strip())
```

Output:

```
First line
Second line
Third line
```

Program 4: Count the number of words in a text file

Code:

```
# Count words using split()
with open("words.txt", "w") as file:
    file.write("Python is easy to learn")

with open("words.txt", "r") as file:
    words = file.read().split()

print("Number of words:", len(words))
```

Output:

```
Number of words: 5
```

Program 5: Count the number of lines and characters in a file

Code:

```
# Count lines and characters
with open("count.txt", "w") as file:
    file.write("Hello\\nPython")

with open("count.txt", "r") as file:
    data = file.read()

print("Number of lines:", len(data.splitlines()))
print("Number of characters:", len(data))
```

Output:

```
Number of lines: 2
Number of characters: 12
```

Program 6: Display only even-numbered lines from a file

Code:

```
# Display lines numbered 2, 4, 6, ...
with open("even_lines.txt", "w") as file:
    file.write("One\\nTwo\\nThree\\nFour")

with open("even_lines.txt", "r") as file:
    for number, line in enumerate(file, start=1):
        if number % 2 == 0:
            print(line.strip())
```

Output:

```
Two
Four
```

Program 7: Write multiple lines of input from the user into a file

Code:

```
# Enter lines; type 'stop' to finish
with open("user_lines.txt", "w") as file:
    while True:
        line = input("Enter text (stop to finish): ")
        if line.lower() == "stop":
            break
        file.write(line + "\\n")

print("Lines saved successfully.")
```

Output:

```
Enter text (stop to finish): Good morning
Enter text (stop to finish): Python lab
Enter text (stop to finish): stop
Lines saved successfully.
```

Program 8: Append new content to an existing file

Code:

```
# Add content without deleting old content
with open("append.txt", "w") as file:
    file.write("First line\\n")

with open("append.txt", "a") as file:
    file.write("New line added")

with open("append.txt", "r") as file:
    print(file.read(), end="")
```

Output:

```
First line
New line added
```

Program 9: Copy the contents of one file into another

Code:

```
# Copy source file into destination file
with open("source.txt", "w") as file:
    file.write("This is copied text.")

with open("source.txt", "r") as source:
    data = source.read()

with open("destination.txt", "w") as destination:
    destination.write(data)

print("File copied successfully.")
```

Output:

```
File copied successfully.
```

Program 10: Read a file and convert its content to uppercase

Code:

```
# Convert all text to uppercase
with open("upper.txt", "w") as file:
    file.write("Python file handling")

with open("upper.txt", "r") as file:
    text = file.read().upper()

print(text)
```

Output:

```
PYTHON FILE HANDLING
```

Program 11: Search for a particular word in a file

Code:

```
# Check whether a word is present
with open("search.txt", "w") as file:
    file.write("Python is a popular language")

word = "popular"
with open("search.txt", "r") as file:
    if word in file.read().split():
        print("Word found")
    else:
        print("Word not found")
```

Output:

```
Word found
```

Program 12: Count the occurrence of a specific word in a file

Code:

```
# Count how many times a word appears
with open("occur.txt", "w") as file:
    file.write("Python is easy and Python is powerful")

word = "Python"
with open("occur.txt", "r") as file:
    count = file.read().split().count(word)

print("Occurrences:", count)
```

Output:

```
Occurrences: 2
```

Program 13: Read numbers from a file and display their sum and average

Code:

```
# Read numbers and calculate sum and average
with open("numbers.txt", "w") as file:
    file.write("10 20 30 40")

with open("numbers.txt", "r") as file:
    numbers = list(map(int, file.read().split()))

print("Sum:", sum(numbers))
print("Average:", sum(numbers) / len(numbers))
```

Output:

```
Sum: 100
Average: 25.0
```

Program 14: Display the longest line in a file

Code:

```
# Find the line having maximum length
with open("longest.txt", "w") as file:
    file.write("Short line\\nThis is the longest line\\nMedium")

with open("longest.txt", "r") as file:
    longest = max(file, key=len)

print("Longest line:", longest.strip())
```

Output:

```
Longest line: This is the longest line
```

Program 15: Replace a word in a file with another word

Code:

```
# Replace old word with new word
with open("replace.txt", "w") as file:
    file.write("I like Java")

with open("replace.txt", "r") as file:
    text = file.read()

text = text.replace("Java", "Python")

with open("replace.txt", "w") as file:
    file.write(text)

print(text)
```

Output:

```
I like Python
```

Program 16: Display the first and last lines of a file

Code:

```
# Display first and last lines
with open("first_last.txt", "w") as file:
    file.write("First line\\nMiddle line\\nLast line")

with open("first_last.txt", "r") as file:
    lines = file.readlines()

print("First:", lines[0].strip())
print("Last:", lines[-1].strip())
```

Output:

```
First: First line
Last: Last line
```

Program 17: Remove blank lines from a file

Code:

```
# Keep only non-blank lines
with open("blank.txt", "w") as file:
    file.write("Apple\\n\\nBanana\\n\\nMango")

with open("blank.txt", "r") as file:
    lines = file.readlines()

with open("clean.txt", "w") as file:
    for line in lines:
        if line.strip():
            file.write(line)

with open("clean.txt", "r") as file:
    print(file.read(), end="")
```

Output:

```
Apple
Banana
Mango
```

Program 18: Read a file and store only unique words into another file

Code:

```
# Remove repeated words while keeping original order
with open("repeat.txt", "w") as file:
    file.write("apple mango apple banana mango")

with open("repeat.txt", "r") as file:
    words = file.read().split()

unique_words = list(dict.fromkeys(words))

with open("unique.txt", "w") as file:
    file.write(" ".join(unique_words))

print("Unique words:", " ".join(unique_words))
```

Output:

```
Unique words: apple mango banana
```

Program 19: Sort all words in a file alphabetically and save to a new file

Code:

```
# Sort words and save them
with open("sort.txt", "w") as file:
    file.write("banana apple mango")

with open("sort.txt", "r") as file:
    words = file.read().split()

words.sort()

with open("sorted.txt", "w") as file:
    file.write(" ".join(words))

print("Sorted words:", " ".join(words))
```

Output:

```
Sorted words: apple banana mango
```

Program 20: Merge two text files into a third file

Code:

```
# Combine contents of two files
with open("one.txt", "w") as file:
    file.write("Hello ")

with open("two.txt", "w") as file:
    file.write("World")

with open("one.txt", "r") as first, open("two.txt", "r") as second:
    merged = first.read() + second.read()

with open("merged.txt", "w") as file:
    file.write(merged)

print("Merged content:", merged)
```

Output:

```
Merged content: Hello World
```

Program 21: Create and write data into a CSV file

Code:

```
# Create a CSV file using csv module
import csv

rows = [["Name", "Marks"], ["Aman", 85], ["Riya", 92]]

with open("students.csv", "w", newline="") as file:
    writer = csv.writer(file)
    writer.writerows(rows)

print("CSV file created successfully.")
```

Output:

```
CSV file created successfully.
```

Program 22: Read data from a CSV file and display it

Code:

```
# Read and display CSV rows
import csv

with open("students.csv", "w", newline="") as file:
    writer = csv.writer(file)
    writer.writerows([["Name", "Marks"], ["Aman", 85], ["Riya", 92]])

with open("students.csv", "r") as file:
    reader = csv.reader(file)
    for row in reader:
        print(row)
```

Output:

```
['Name', 'Marks']
['Aman', '85']
['Riya', '92']
```

Program 23: Read a CSV file and calculate the average of numeric columns

Code:

```
# Calculate average marks from CSV
import csv

with open("marks.csv", "w", newline="") as file:
    writer = csv.writer(file)
    writer.writerow(["Name", "Marks"])
    writer.writerows([["Aman", 80], ["Riya", 90], ["Neha", 70]])

with open("marks.csv", "r") as file:
    reader = csv.DictReader(file)
    marks = [int(row["Marks"]) for row in reader]

print("Average marks:", sum(marks) / len(marks))
```

Output:

```
Average marks: 80.0
```

Program 24: Count the number of rows and columns in a CSV file

Code:

```
# Count CSV rows and columns (excluding header)
import csv

with open("table.csv", "w", newline="") as file:
    writer = csv.writer(file)
    writer.writerows([["Name", "Age"], ["Aman", 20], ["Riya", 21]])

with open("table.csv", "r") as file:
    rows = list(csv.reader(file))

print("Number of rows:", len(rows) - 1)
print("Number of columns:", len(rows[0]))
```

Output:

```
Number of rows: 2
Number of columns: 2
```

Program 25: Append a new record to an existing CSV file

Code:

```
# Add one more student record
import csv

with open("append.csv", "w", newline="") as file:
    writer = csv.writer(file)
    writer.writerow(["Name", "Marks"])
    writer.writerow(["Aman", 85])

with open("append.csv", "a", newline="") as file:
    writer = csv.writer(file)
    writer.writerow(["Riya", 92])

print("New record added.")
```

Output:

```
New record added.
```

Program 26: Copy data from one CSV file to another

Code:

```
# Copy all CSV rows
import csv

with open("source.csv", "w", newline="") as file:
    csv.writer(file).writerows([["Name", "Marks"], ["Aman", 85]])

with open("source.csv", "r") as source:
    rows = list(csv.reader(source))

with open("copy.csv", "w", newline="") as destination:
    csv.writer(destination).writerows(rows)

print("CSV data copied successfully.")
```

Output:

```
CSV data copied successfully.
```

Program 27: Display specific columns (Name and Marks) from a CSV file

Code:

```
# Display selected CSV columns
import csv

with open("students2.csv", "w", newline="") as file:
    writer = csv.writer(file)
    writer.writerow(["Name", "Age", "Marks"])
    writer.writerows([["Aman", 20, 85], ["Riya", 21, 92]])

with open("students2.csv", "r") as file:
    reader = csv.DictReader(file)
    for row in reader:
        print(row["Name"], row["Marks"])
```

Output:

```
Aman 85
Riya 92
```

Program 28: Find and display the highest marks from a CSV file

Code:

```
# Find the student with highest marks
import csv

with open("highest.csv", "w", newline="") as file:
    writer = csv.writer(file)
    writer.writerow(["Name", "Marks"])
    writer.writerows([["Aman", 85], ["Riya", 92], ["Neha", 78]])

with open("highest.csv", "r") as file:
    reader = csv.DictReader(file)
    top = max(reader, key=lambda row: int(row["Marks"]))

print("Highest:", top["Name"], top["Marks"])
```

Output:

```
Highest: Riya 92
```

Program 29: Search a record in a CSV file based on student name

Code:

```
# Search for a student by name
import csv

with open("find.csv", "w", newline="") as file:
    writer = csv.writer(file)
    writer.writerow(["Name", "Marks"])
    writer.writerows([["Aman", 85], ["Riya", 92]])

search_name = "Riya"
with open("find.csv", "r") as file:
    reader = csv.DictReader(file)
    for row in reader:
        if row["Name"].lower() == search_name.lower():
            print("Record found:", row["Name"], row["Marks"])
            break
    else:
        print("Record not found")
```

Output:

```
Record found: Riya 92
```

Program 30: Combine multiple CSV files into one single file

Code:

```
# Merge rows from two CSV files
import csv

data1 = [["Name", "Marks"], ["Aman", 85]]
data2 = [["Name", "Marks"], ["Riya", 92]]

with open("class1.csv", "w", newline="") as file:
    csv.writer(file).writerows(data1)

with open("class2.csv", "w", newline="") as file:
    csv.writer(file).writerows(data2)

combined = []
for filename in ["class1.csv", "class2.csv"]:
    with open(filename, "r") as file:
        rows = list(csv.reader(file))
        combined.extend(rows[1:])  # Skip repeated header

with open("all_students.csv", "w", newline="") as file:
    writer = csv.writer(file)
    writer.writerow(["Name", "Marks"])
    writer.writerows(combined)

print("All CSV files combined successfully.")
print("Total records:", len(combined))
```

Output:

```
All CSV files combined successfully.
Total records: 2
```